

|               |                                         |
|---------------|-----------------------------------------|
| Date          | 15/07/2026                              |
| Team ID       | xxxxxxx                                 |
| Project Name  | Smart Lender (Loan Approval Prediction) |
| Maximum Marks | 5 Marks                                 |

Code-Layout, Readability Template

Evaluation of project code quality, structure, and modular components.

Code Quality Checklist

| S.N<br>o | Parameter           | Description                                                           | Followed | Remarks                     |
|----------|---------------------|-----------------------------------------------------------------------|----------|-----------------------------|
| 1        | Indentation         | Consistent use of 4 spaces per indentation level.                     | Yes      | PEP8 compliant.             |
| 2        | Directory Structure | Organized into Dataset/, notebooks/, models/, and templates/ folders. | Yes      | Matches Flask structure.    |
| 3        | Variable Names      | Descriptive names (e.g. input_feature, data_scaled, prediction).      | Yes      | Enhances code readability.  |
| 4        | Comments            | Explanation of preprocessing, mapping, scaling, and inference.        | Yes      | Inline docstrings added.    |
| 5        | Modular Logic       | Separation of model training (.ipynb) and web inference (app.py).     | Yes      | Clean boundary.             |
| 6        | Error Handling      | Input type casting to float and exception fallback mapping.           | Yes      | Prevents Flask 500 crashes. |

## Reusable Components

| Component Name          | Technology     | Location Used                       | Reusability Level |
|-------------------------|----------------|-------------------------------------|-------------------|
| Dynamic Path Resolver   | Python os.path | app.py model and scaler loading     | High              |
| Categorical Data Mapper | Python dict    | app.py preprocessing data pipelines | High              |
| StandardScaler Model    | Scikit-learn   | app.py input feature scaling        | High              |